

Advanced Time Series PSET 3

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Question 1

The NKPC is

$$\pi_t = c + \gamma_f E_t(\pi_{t+1}) + \gamma_b \pi_{t-1} + \lambda x_t + \nu_t, \quad \gamma_b = 1 - \gamma_f,$$

and inflation is assumed to satisfy

$$\pi_t = \mu_\pi + \rho_\pi \pi_{t-1} + \rho_x x_{t-1} + \eta_t, \quad (\nu_t, \eta_t)' \stackrel{i.i.d.}{\sim} (0, \Sigma).$$

We measure inflation as

$$\pi_t = 100\Delta \log(\text{GDPDEF}_t),$$

the quarter-over-quarter log change in the GDP deflator in percentage points.

(i)

Since the information set at date t contains current and lagged (π_t, x_t) , rational expectations and the inflation ADL imply

$$E_t(\pi_{t+1}) = \mu_\pi + \rho_\pi \pi_t + \rho_x x_t.$$

Substituting this into the NKPC gives

$$\pi_t = c + \gamma_f(\mu_\pi + \rho_\pi \pi_t + \rho_x x_t) + (1 - \gamma_f)\pi_{t-1} + \lambda x_t + \nu_t.$$

Let

$$d(\theta) \equiv \lambda + \gamma_f \rho_x.$$

When $d(\theta) \neq 0$, rearranging the NKPC gives

$$x_t = \frac{\mu_\pi(1 - \gamma_f \rho_\pi - \gamma_f) - c}{d(\theta)} + \frac{(1 - \gamma_f \rho_\pi)\rho_\pi - (1 - \gamma_f)}{d(\theta)} \pi_{t-1} + \frac{(1 - \gamma_f \rho_\pi)\rho_x}{d(\theta)} x_{t-1} + \frac{(1 - \gamma_f \rho_\pi)\eta_t - \nu_t}{d(\theta)}.$$

Combining this equation with the ADL equation for π_t , the vector

$$y_t \equiv (\pi_t, x_t)'$$

follows a bivariate VAR(1),

$$y_t = a + A(\theta)y_{t-1} + u_t,$$

where the four slope coefficients are

$$A(\theta) = \begin{pmatrix} \rho_\pi & \rho_x \\ \frac{(1 - \gamma_f \rho_\pi)\rho_\pi - (1 - \gamma_f)}{\lambda + \gamma_f \rho_x} & \frac{(1 - \gamma_f \rho_\pi)\rho_x}{\lambda + \gamma_f \rho_x} \end{pmatrix}.$$

The intercepts and innovations are

$$a = \begin{pmatrix} \mu_\pi \\ \frac{\mu_\pi(1 - \gamma_f \rho_\pi - \gamma_f) - c}{\lambda + \gamma_f \rho_x} \end{pmatrix}, \quad u_t = \begin{pmatrix} \eta_t \\ \frac{(1 - \gamma_f \rho_\pi)\eta_t - \nu_t}{\lambda + \gamma_f \rho_x} \end{pmatrix}.$$

Because $(\nu_t, \eta_t)'$ is i.i.d., u_t is also an i.i.d. reduced-form innovation.

(ii)

We estimate the unrestricted reduced-form VAR(1)

$$y_t = a + Ay_{t-1} + u_t$$

by least squares using the same source data and sample as PSET 2, Question 6: GDPDEF and quarterly-average UNRATE, restricted to the largest overlapping sample before 2020 and using the percentage-point inflation scaling stated above. The raw sample is 1948Q1–2019Q4 (288 quarters), and the VAR regression sample is 1948Q2–2019Q4 (287 observations).

The estimated intercepts are

$$\hat{a}_\pi = 0.121547, \quad \hat{a}_x = 0.1222.$$

The estimated slope matrix is

$$\hat{A} = \begin{pmatrix} 0.7856 & 0.006918 \\ 0.0531 & 0.9716 \end{pmatrix},$$

with rows corresponding to the π_t and x_t equations and columns corresponding to π_{t-1} and x_{t-1} . The largest absolute eigenvalue of $\hat{A}$ is 0.9736.

The minimum-distance estimator uses the restriction

$$\hat{\theta} = \arg \min_{\theta} \{\text{vec}(\hat{A}) - \text{vec}(A(\theta))\}' W \{\text{vec}(\hat{A}) - \text{vec}(A(\theta))\}.$$

We use $W = I_4$. Since the model gives four reduced-form slope coefficients for the four parameters $\theta = (\gamma_f, \lambda, \rho_\pi, \rho_x)'$, the estimate is exactly identified. In particular,

$$\hat{\rho}_\pi = \hat{A}_{11}, \quad \hat{\rho}_x = \hat{A}_{12},$$

and, using the last two VAR coefficients,

$$\hat{\gamma}_f = \frac{\hat{A}_{22}(1 - \hat{\rho}_\pi) + \hat{A}_{21}\hat{\rho}_x}{\hat{A}_{21}\hat{\rho}_\pi\hat{\rho}_x + \hat{A}_{22}(1 - \hat{\rho}_\pi^2)},$$

$$\hat{\lambda} = \frac{(1 - \hat{\gamma}_f\hat{\rho}_\pi)\hat{\rho}_x}{\hat{A}_{22}} - \hat{\gamma}_f\hat{\rho}_x.$$

The resulting estimates are:

Parameter	Estimate	95% bootstrap lower	95% bootstrap upper
γ_f	0.560604	0.544364	0.593460
λ	1.063×10^{-4}	-0.001254	0.001331
ρ_π	0.785551	0.686598	0.836473
ρ_x	0.006918	-0.032102	0.040942

(iii)

We compute the confidence intervals using a recursive residual bootstrap for the fitted VAR(1). The procedure is:

1. Estimate the VAR(1) by OLS and collect the centered residual vectors $\hat{u}_t - \bar{\hat{u}}$.
2. For each bootstrap draw $b = 1, \dots, 10000$, sample residual vectors with replacement from the centered empirical residual distribution, preserving the cross-equation dependence between the π_t and x_t shocks.
3. Starting from the observed initial value $y_0^* = y_0$, recursively generate

$$y_t^* = \hat{a} + \hat{A}y_{t-1}^* + u_t^*.$$

4. Re-estimate the VAR(1) on the bootstrap sample and recompute $\hat{\theta}^* = (\hat{\gamma}_f^*, \hat{\lambda}^*, \hat{\rho}_\pi^*, \hat{\rho}_x^*)'$ using the same minimum-distance inversion.
5. Report the 2.5th and 97.5th percentiles of the bootstrap estimates.

We use 10000 bootstrap repetitions with random seed 31730; all 10000 repetitions produced non-singular estimates. This bootstrap relies on the VAR(1) restriction derived above, stationarity of the fitted VAR, and the i.i.d. innovation assumption. The i.i.d. assumption is exactly the extra structure imposed in the question, since the reduced-form VAR innovations are linear transformations of $(\nu_t, \eta_t)'$.

Question 2

We observe

$$y_t = \mu + z_t,$$

where μ is unknown and $\{z_t\}$ is mean zero, strictly stationary, and weakly dependent. The sample mean is

$$\hat{\mu} = \frac{1}{T} \sum_{t=1}^T y_t.$$

The long-run variance needed for inference on μ is

$$\Omega = \text{LRV}(z_t) = \sum_{\ell=-\infty}^{\infty} \Gamma_z(\ell).$$

Since y_t differs from z_t only by a constant, the population autocovariances of $\{y_t\}$ and $\{z_t\}$ are the same. The kernel HAC estimator is

$$\hat{\Omega} = \sum_{\ell=-(T-1)}^{T-1} k\left(\frac{\ell}{S_T}\right) \hat{\Gamma}(\ell),$$

where, for $\ell \geq 0$,

$$\hat{\Gamma}(\ell) = \frac{1}{T} \sum_{t=\ell+1}^T (y_t - \hat{\mu})(y_{t-\ell} - \hat{\mu}), \quad \hat{\Gamma}(-\ell) = \hat{\Gamma}(\ell)$$

in this scalar problem.

Under the Andrews assumptions, the one-dimensional MSE-optimal bandwidth is

$$S_T^* = \left[\frac{q^* k_{q^*}^2}{\int_{-\infty}^{\infty} k(x)^2 dx} \alpha(q^*) T \right]^{1/(2q^*+1)},$$

where

$$\alpha(q) = \left(\frac{s_z^{(q)}(0)}{\Omega} \right)^2, \quad s_z^{(q)}(0) = \sum_{\ell=-\infty}^{\infty} |\ell|^q \Gamma_z(\ell).$$

(i)

Under the stationary AR(1) working model

$$z_t = \rho z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim WN(0, \sigma^2), \quad |\rho| < 1.$$

Its autocovariance function is

$$\Gamma_z(\ell) = \frac{\sigma^2}{1 - \rho^2} \rho^{|\ell|}.$$

Therefore

$$\Omega = \frac{\sigma^2}{1 - \rho^2} \left(1 + 2 \sum_{\ell=1}^{\infty} \rho^\ell \right) = \frac{\sigma^2}{1 - \rho^2} \left(\frac{1 + \rho}{1 - \rho} \right) = \frac{\sigma^2}{(1 - \rho)^2}.$$

For any $q > 0$, the zero-lag term in $s_z^{(q)}(0)$ is zero, and since $|\rho| < 1$ the following series is absolutely convergent:

$$s_z^{(q)}(0) = 2 \sum_{\ell=1}^{\infty} \ell^q \Gamma_z(\ell) = \frac{2\sigma^2}{1 - \rho^2} \sum_{\ell=1}^{\infty} \ell^q \rho^\ell.$$

Thus

$$\frac{s_z^{(q)}(0)}{\Omega} = \frac{2(1 - \rho)}{1 + \rho} \sum_{\ell=1}^{\infty} \ell^q \rho^\ell,$$

and

$$\alpha(q) = \left[\frac{2(1 - \rho)}{1 + \rho} \sum_{\ell=1}^{\infty} \ell^q \rho^\ell \right]^2.$$

Substituting this into Andrews's formula gives

$$S_T^* = \left[\frac{q^* k_{q^*}^2 T}{\int_{-\infty}^{\infty} k(x)^2 dx} \left\{ \frac{2(1 - \rho)}{1 + \rho} \sum_{\ell=1}^{\infty} \ell^{q^*} \rho^\ell \right\}^2 \right]^{1/(2q^*+1)}.$$

This depends only on T , ρ , and the chosen kernel. The innovation variance σ^2 cancels, as it should: the plug-in bandwidth is determined by persistence, not by the scale of the series.

(ii)

The Bartlett kernel is

$$k(x) = \max\{1 - |x|, 0\}.$$

Near zero, $k(x) = 1 - |x|$, so

$$k_1 = \lim_{x \rightarrow 0} \frac{1 - k(x)}{|x|} = 1.$$

For $q > 1$,

$$\frac{1 - k(x)}{|x|^q} = |x|^{1-q} \rightarrow \infty,$$

so the characteristic exponent is $q^* = 1$. Also

$$\int_{-\infty}^{\infty} k(x)^2 dx = \int_{-1}^1 (1 - |x|)^2 dx = 2 \int_0^1 (1 - x)^2 dx = \frac{2}{3}.$$

For $q = 1$,

$$\sum_{\ell=1}^{\infty} \ell \rho^{\ell} = \frac{\rho}{(1 - \rho)^2},$$

so

$$\frac{s_z^{(1)}(0)}{\Omega} = \frac{2(1 - \rho)}{1 + \rho} \frac{\rho}{(1 - \rho)^2} = \frac{2\rho}{1 - \rho^2},$$

and

$$\alpha(1) = \frac{4\rho^2}{(1 - \rho^2)^2}.$$

Substituting $q^* = 1$, $k_1 = 1$, and $\int k^2 = 2/3$ into the optimal bandwidth formula gives

$$S_{T,\text{Bartlett}}^* = \left[\frac{1}{2/3} \frac{4\rho^2}{(1 - \rho^2)^2} T \right]^{1/3} = \left[\frac{6\rho^2}{(1 - \rho^2)^2} T \right]^{1/3}.$$

Equivalently,

$$S_{T,\text{Bartlett}}^* = \left(\frac{6\rho^2}{(1 - \rho^2)^2} \right)^{1/3} T^{1/3}.$$

For example, if $\rho = 0.25$, this gives approximately $0.75T^{1/3}$, matching the standard rule of thumb.

For the quadratic spectral kernel,

$$k(x) = \frac{25}{12\pi^2 x^2} \left[\frac{\sin(6\pi x/5)}{6\pi x/5} - \cos(6\pi x/5) \right], \quad x \neq 0,$$

with $k(0) = 1$. The problem states that

$$\int_{-\infty}^{\infty} k(x)^2 dx = 1, \quad q^* = 2, \quad k_2 = 1.421223.$$

To verify the local expansion, set $a = 6\pi/5$. Since

$$\frac{\sin(ax)}{ax} = 1 - \frac{a^2 x^2}{6} + \frac{a^4 x^4}{120} + O(x^6), \quad \cos(ax) = 1 - \frac{a^2 x^2}{2} + \frac{a^4 x^4}{24} + O(x^6),$$

we have

$$\frac{\sin(ax)}{ax} - \cos(ax) = \frac{a^2 x^2}{3} - \frac{a^4 x^4}{30} + O(x^6).$$

Multiplying by $25/(12\pi^2 x^2)$ gives

$$k(x) = 1 - \frac{18\pi^2}{125} x^2 + O(x^4),$$

so

$$k_2 = \frac{18\pi^2}{125} = 1.421223\dots, \quad q^* = 2.$$

For $q = 2$,

$$\sum_{\ell=1}^{\infty} \ell^2 \rho^\ell = \frac{\rho(1+\rho)}{(1-\rho)^3},$$

and therefore

$$\frac{s_z^{(2)}(0)}{\Omega} = \frac{2(1-\rho)}{1+\rho} \frac{\rho(1+\rho)}{(1-\rho)^3} = \frac{2\rho}{(1-\rho)^2}.$$

Thus

$$\alpha(2) = \frac{4\rho^2}{(1-\rho)^4}.$$

Substituting $q^* = 2$, $k_2 = 1.421223$, and $\int k^2 = 1$,

$$S_{T, \text{QS}}^* = \left[2(1.421223)^2 \frac{4\rho^2}{(1-\rho)^4} T \right]^{1/5} = \left[\frac{8(1.421223)^2 \rho^2}{(1-\rho)^4} T \right]^{1/5}.$$

Equivalently,

$$S_{T, \text{QS}}^* = 1.3221 \left[\frac{4\rho^2}{(1-\rho)^4} T \right]^{1/5}, \quad 1.3221 = \{2(1.421223)^2\}^{1/5}.$$

The difference in rates is coming from the kernel smoothness at the origin: Bartlett has $q^* = 1$, so its bandwidth is proportional to $T^{1/3}$, while QS has $q^* = 2$, so its bandwidth is proportional to $T^{1/5}$. The denominators also line up with the AR(1) autocovariances: Bartlett depends on $1 - \rho^2$ because $s_z^{(1)}(0)/\Omega = 2\rho/(1 - \rho^2)$, whereas QS depends on $(1 - \rho)^4$ because $s_z^{(2)}(0)/\Omega = 2\rho/(1 - \rho)^2$. If $\rho = 0$, the leading bias term in the Andrews expansion is zero, so the plug-in formulas return $S_T^* = 0$. In practice we would impose an integer bandwidth rule, but this white-noise edge case is a feature of the plug-in MSE calculation rather than an algebraic problem.

Question 3

We collect the four macro-financial variables in

$$y_t = (\text{ipgr}_t, \text{infl}_t, \text{gs1}_t, \text{ebp}_t)'$$

The external instrument is $z_t = \text{ff4.tc}_t$, the high-frequency change in three-month-ahead federal funds futures prices around FOMC announcements. This is the same type of high-frequency policy surprise emphasized by Gertler and Karadi (2015): it uses variation in futures prices in a narrow policy-announcement window as an external source of monetary policy variation, avoiding the simultaneity problem that arises when financial variables enter the VAR. Following the question, we include an intercept and 4 lags in every VAR and regression. The one-year Treasury rate, **gs1**, is the policy indicator, and all responses below are normalized so that the impact response of **gs1** equals 1 (100 basis points).

(i)

We estimate the reduced-form VAR

$$y_t = c + \sum_{j=1}^p A_j y_{t-j} + \eta_t, \quad p = 4,$$

by least squares. Let

$$\Sigma = E(\eta_t \eta_t'), \quad \gamma = E(\eta_t z_t).$$

Under the SVAR-IV assumptions,

$$\eta_t = H \varepsilon_t, \quad E(z_t \varepsilon_{1t}) \neq 0, \quad E(z_t \varepsilon_{jt}) = 0 \quad (j \geq 2),$$

with $E(\varepsilon_t \varepsilon_t') = I$. Therefore $\gamma = h_1 E(z_t \varepsilon_{1t})$, where h_1 is the first column of H , so γ is proportional to the impact vector of the monetary policy shock. With the unit-variance structural-shock normalization,

$$h_1 = \frac{\gamma}{(\gamma' \Sigma^{-1} \gamma)^{1/2}},$$

up to sign. We choose the sign so the **gs1** impact response is positive. The estimated first-stage regression of the **gs1** VAR residual on **ff4.tc** has $F = 16.19$ and $R^2 = 5.78\%$, so the instrument is relevant in this sample.

Let Ψ_h denote the VAR moving-average coefficient matrices, with

$$\Psi_0 = I, \quad \Psi_h = \sum_{j=1}^{\min(p,h)} A_j \Psi_{h-j}.$$

The normalized impulse response is

$$\text{IRF}_h = \Psi_h \frac{h_1}{e'_{\text{gs1}} h_1}, \quad h = 0, 1, \dots, 36.$$

The resulting EBP response is shown in Figure 1. The impact response is 0.711 percentage points. It remains positive for the whole 36-month window, falling to 0.299 after 12 months and 0.077 after 36 months. The sign and magnitude are consistent with the main Gertler-Karadi mechanism: a contractionary policy surprise raises credit costs, and the excess bond premium moves by a large amount relative to the one-year rate normalization.

(ii)

For the EBP forecast variance decomposition, use the unit-variance impact vector h_1 , not the 100-basis-point rescaled vector. For horizon $\ell \geq 1$,

$$\text{FVD}_{\text{ebp},1,\ell} = \frac{\sum_{h=0}^{\ell-1} (e'_{\text{ebp}} \Psi_h h_1)^2}{\sum_{h=0}^{\ell-1} e'_{\text{ebp}} \Psi_h \Sigma \Psi_h' e_{\text{ebp}}}.$$

The numerator is the forecast-error variance contribution of the monetary shock, and the denominator is the total reduced-form VAR forecast-error variance. The estimated FVD is plotted in Figure 2. Monetary policy shocks explain about 18.94% of the one-month EBP forecast-error variance, 18.78% at 12 months, and 21.28% at 36 months.

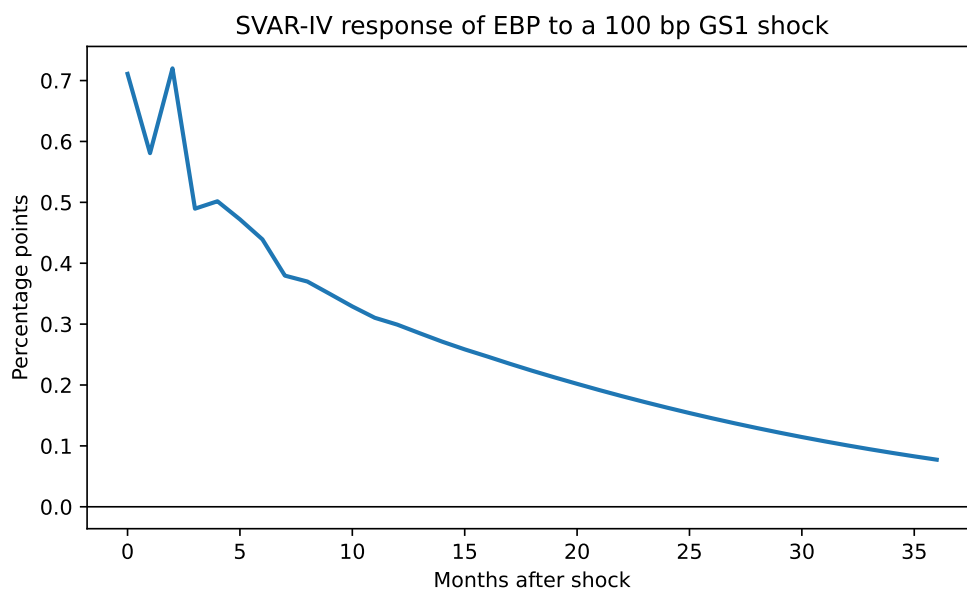


Figure 1: SVAR-IV impulse response of the excess bond premium to a monetary policy shock that raises `gs1` by 100 basis points on impact.

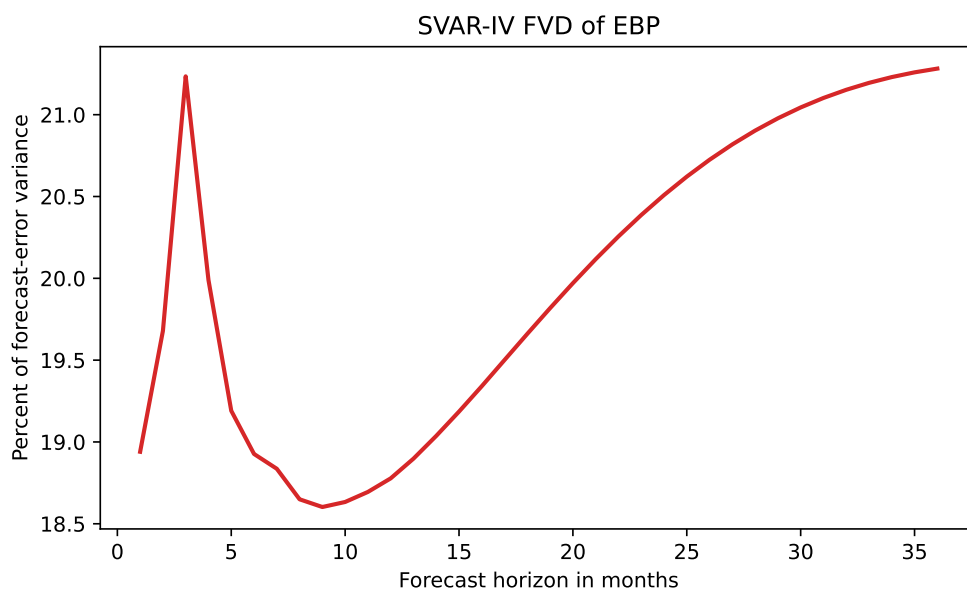


Figure 2: SVAR-IV forecast variance decomposition of the excess bond premium with respect to the monetary policy shock.

(iii)

We next drop the invertibility assumption. Following the standard LP-IV argument, form

$$w_t = (z_t, y'_t)' = (\mathbf{ff4_tc}_t, \mathbf{ipgr}_t, \mathbf{infl}_t, \mathbf{gs1}_t, \mathbf{ebp}_t)'.$$

We estimate a VAR(4) for w_t , order z_t first, and use a recursive Cholesky shock to the first variable. Since z_t is only a noisy proxy for the monetary shock, the absolute scale of this recursive shock is not structural. However, the attenuation from measurement error is common across horizons and variables, so the relative IRF is valid after normalizing by the impact response of $\mathbf{gs1}$. Thus we again divide the responses by the impact response of $\mathbf{gs1}$.

The recursive-VAR LP-IV estimate is close to the SVAR-IV estimate on impact: 0.693 percentage points for EBP. It then decays somewhat faster, reaching 0.198 at 12 months and 0.036 at 36 months. The comparison is shown in Figure 3.

(iv)

The direct 2SLS implementation estimates, for each horizon $h = 0, \dots, 36$,

$$\mathbf{ebp}_{t+h} = \alpha_h + \beta_h \mathbf{gs1}_t + \delta'_h X_{t-1} + u_{t+h|t},$$

using $z_t = \mathbf{ff4_tc}_t$ as an instrument for $\mathbf{gs1}_t$. The control vector X_{t-1} contains an intercept, 4 lags of y_t , and 4 lags of z_t . With a single endogenous regressor and a single IV, the coefficient can be written after residualizing on controls as

$$\hat{\beta}_h = \frac{\tilde{z}'_h \tilde{y}_h}{\tilde{z}'_h \mathbf{gs1}},$$

where tildes denote residuals after projection on X_{t-1} . The first-stage F -statistic at $h = 0$ is 23.49, with partial $R^2 = 8.78\%$.

The 2SLS LP-IV estimates are more jagged than the VAR-based estimates, which is expected because each horizon is estimated separately and no VAR propagation restrictions are imposed. The impact response is mechanically the same as the recursive-VAR LP-IV impact response, 0.693. At longer horizons the direct estimates are larger: 0.845 at 12 months and 0.779 at 36 months.

Horizon	SVAR-IV	LP-IV recursive VAR	LP-IV 2SLS
0	0.711	0.693	0.693
6	0.439	0.395	1.337
12	0.299	0.198	0.845
24	0.163	0.090	0.968
36	0.077	0.036	0.779

(v)

We know the key testable implication of invertibility with an external IV: if the monetary policy shock is invertible with respect to $\{y_t\}$, then the proxy z_t should not Granger-cause y_t conditional on lags of y_t . Intuitively, under invertibility the useful shock component of z_t is already contained in the history of y_t , while the measurement-error component of z_t is not useful for forecasting.

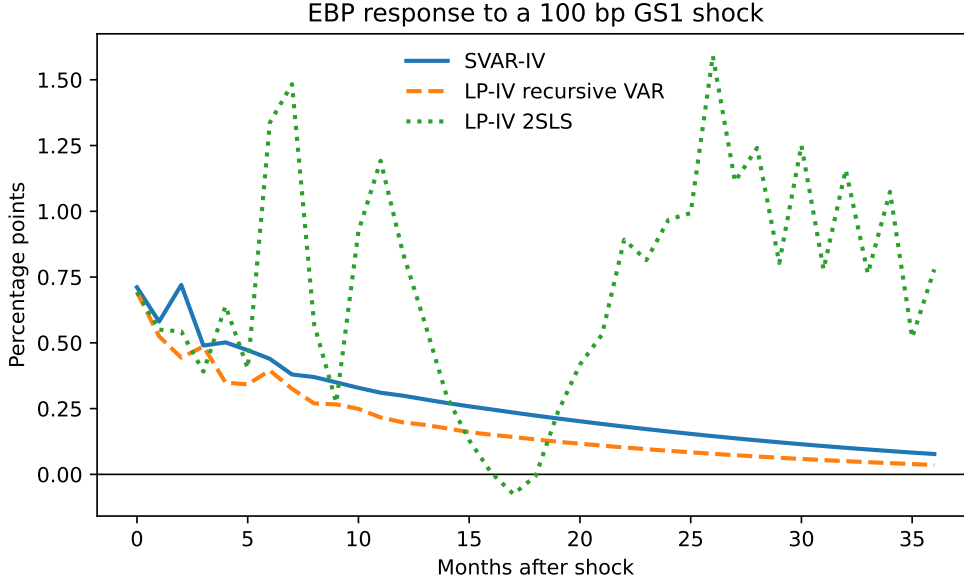


Figure 3: EBP impulse responses from SVAR-IV, recursive-VAR LP-IV, and direct 2SLS LP-IV. All responses are normalized to a 100 basis point impact response of `gs1`.

Using the VAR specification in part (iii), we test the joint null that lags 1 through 4 of `ff4_tc` have zero coefficients in the four y_t equations. This imposes $4 \times 4 = 16$ restrictions. The Gaussian VAR likelihood-ratio statistic is

$$LR = T \left(\log |\hat{\Sigma}_R| - \log |\hat{\Sigma}_U| \right),$$

where $\hat{\Sigma}_U$ is the residual covariance matrix from the unrestricted y_t equations with lags of $(z_t, y_t)'$, and $\hat{\Sigma}_R$ is from the restricted equations with only lags of y_t . The result is

$$LR = 14.18, \quad df = 16, \quad p\text{-value} = 0.585.$$

Thus we do not reject the no-Granger-causality implication of invertibility. This does not prove invertibility, but it means this particular testable implication is not contradicted by this VAR specification.

(vi)

We compute confidence intervals by a recursive residual bootstrap for the joint VAR in

$$w_t = (z_t, y_t)'$$

This is the reduced-form object used in the recursive-VAR LP-IV estimator, and it also gives the time-series model for z_t that is needed when bootstrapping SVAR-IV with an external instrument. After estimating the VAR(4) for w_t , we center the joint residuals, resample the residual vectors with replacement, hold the first 4 observations fixed, and recursively generate

$$w_t^* = \hat{c} + \sum_{j=1}^4 \hat{A}_j w_{t-j}^* + u_t^*.$$

For each artificial sample, we re-estimate the SVAR-IV IRF, the SVAR-IV FVD, the recursive-VAR LP-IV IRF, and the direct 2SLS LP-IV IRF exactly as above. This means the bootstrap includes

uncertainty about the VAR propagation, the reduced-form covariance matrix, the proxy covariance

$$\hat{\gamma} = \frac{1}{T} \sum_t \hat{\eta}_t z_t,$$

and the first-stage relationship used in the 2SLS local projections.

For a scalar object θ_h , we report the equal-tailed percentile interval

$$[q_{0.05,h}^*, q_{0.95,h}^*],$$

where $q_{\alpha,h}^*$ is the α -quantile of the bootstrap estimates $\hat{\theta}_h^*$. We use 2,000 bootstrap repetitions with random seed 317303, and all 2,000 repetitions produced finite estimates. The validity of this procedure relies on the maintained finite-order VAR approximation for w_t , stable dynamics, and i.i.d. joint VAR innovations. It is the standard strong-instrument bootstrap approximation rather than a weak-IV-robust procedure; the first-stage statistics above are why this is a reasonable approximation here.

The selected 90% confidence intervals for the EBP impulse responses are:

Estimator	$h = 0$	$h = 12$	$h = 36$
SVAR-IV	0.711 [0.190, 1.826]	0.299 [0.026, 0.702]	0.077 [−0.031, 0.177]
LP-IV recursive VAR	0.693 [0.190, 1.741]	0.198 [−0.260, 0.706]	0.036 [−0.113, 0.167]
LP-IV 2SLS	0.693 [0.190, 1.741]	0.845 [−1.145, 1.493]	0.779 [−1.437, 1.333]

For the EBP forecast variance decomposition, the selected 90% confidence intervals, reported in percent, are:

	$\ell = 1$	$\ell = 12$	$\ell = 36$
SVAR-IV FVD	18.94 [1.86, 48.04]	18.78 [2.70, 48.27]	21.28 [4.57, 49.63]

Figure 4 plots the point estimates and intervals across all horizons. The direct 2SLS intervals are much wider than the two VAR-based intervals, especially at medium and long horizons. That is the empirical cost of estimating each horizon separately without imposing the VAR propagation restrictions. The SVAR-IV and recursive-VAR LP-IV bands are tighter, but they are also more model-dependent because their horizon profiles come from the estimated VAR law of motion.

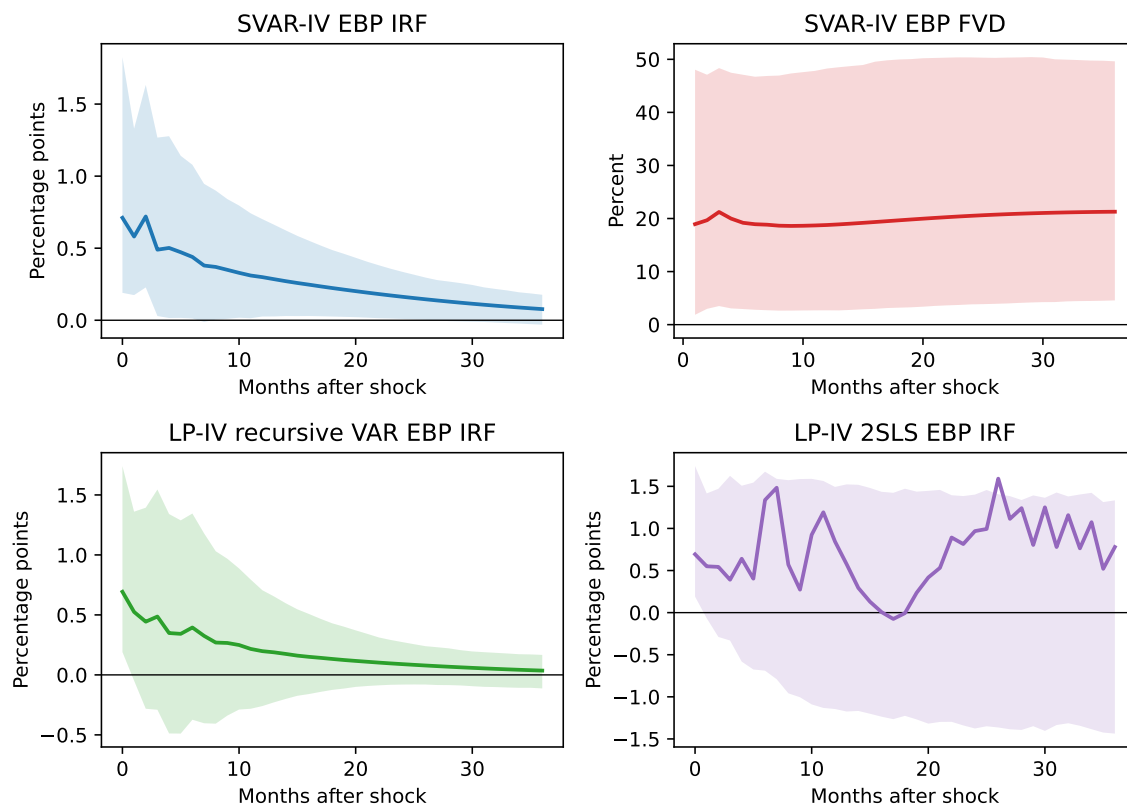


Figure 4: Point estimates and 90% recursive-residual bootstrap confidence intervals for the SVAR-IV IRF, SVAR-IV FVD, recursive-VAR LP-IV IRF, and direct 2SLS LP-IV IRF.

Question 4

We use the FRED series `INDPRO`, `PCEPI`, and `GS1`. Output growth and inflation are monthly log growth rates in annualized percentage points,

$$g_t = 1200\Delta \log(\text{INDPRO}_t), \quad \pi_t = 1200\Delta \log(\text{PCEPI}_t),$$

and the real interest rate is

$$r_t = \text{GS1}_t - \pi_t.$$

After constructing these variables, we keep observations dated 1984M1–2019M12, so the VAR sample after 12 lags is 1985M1–2019M12 (420 observations). Let

$$y_t = (g_t, \pi_t, r_t)'$$

We estimate the reduced-form VAR(12)

$$y_t = c + \sum_{j=1}^{12} A_j y_{t-j} + \eta_t, \quad E(\eta_t \eta_t') = \Sigma,$$

by least squares. The largest absolute eigenvalue of the companion matrix is 0.9786, so the estimated propagation is stable. Using the standard state-space notation, define the companion state

$$s_t = (y_t', y_{t-1}', \dots, y_{t-12+1}')'$$

and write deviations from the deterministic intercept as

$$s_t = F s_{t-1} + G \eta_t, \quad y_t = J s_t.$$

Here F contains $A_1, \dots, A_{12}$ in its first block row and identity matrices below the diagonal, $G = (I_3, 0, \dots, 0)'$, and $J = (I_3, 0, \dots, 0)$. Thus the reduced-form moving-average matrices are

$$\Psi_\ell = J F^\ell G, \quad \ell = 0, 1, \dots, 36.$$

This is the standard linear state-space state-transition logic: an impulse to the innovation changes the state through G , propagates through F^ℓ , and is read off by J .

Let C be the lower Cholesky factor of $\hat{\Sigma}$, so $CC' = \hat{\Sigma}$. Every exactly identified orthogonalization can be written as

$$\eta_t = CQ\varepsilon_t, \quad QQ' = I_3, \quad E(\varepsilon_t \varepsilon_t') = I_3.$$

For a candidate first column $q = Qe_1$, the impulse response to the first unit-variance structural shock is

$$\theta_\ell(q) = e_g' \Psi_\ell Cq, \quad \ell = 0, 1, \dots, 36,$$

where e_g selects output growth. The contractionary monetary shock is restricted to raise the real rate and lower inflation at horizons 0, 1, 2:

$$e_r' \Psi_h Cq \geq 0, \quad e_\pi' \Psi_h Cq \leq 0, \quad h = 0, 1, 2.$$

There is no sign restriction on output growth and no restriction on the other two shocks. Because Haar measure is invariant to relabeling columns, restricting the first column is without loss of generality for the one-shock object. We draw 1,000,000 orthogonal matrices from Haar measure using QR decompositions of Gaussian matrices, normalizing the diagonal of R to be positive as required in the assignment. Of these, 240,577 draws satisfy the restrictions, for an acceptance rate of 24.06%.

(i)

The estimated identified-set bounds for θ_ℓ are the pointwise minimum and maximum values of $\theta_\ell(q)$ across the retained rotations. Selected horizons are:

Horizon	Lower bound	Upper bound	Midpoint	Posterior median
0	-6.063	2.657	-1.703	-2.762
6	-1.373	1.325	-0.024	-0.008
12	-0.043	0.764	0.360	0.521
24	-0.074	0.075	0.001	-0.005
36	-0.035	0.068	0.016	0.019

On impact, the output-growth response is only partially pinned down: $[-6.063, 2.657]$. This wide interval is exactly the point of sign-restriction identification: many different rotations are observationally equivalent and satisfy the same qualitative monetary-policy restrictions.

(ii)

For the Bayesian thought experiment, we treat the reduced-form VAR estimates as fixed and interpret Haar measure, truncated to the sign-restricted set, as the posterior over Q . The posterior median and equal-tailed 68% credible set are therefore the 50th, 16th, and 84th percentiles of the retained $\theta_\ell(q)$ draws. At the same selected horizons:

Horizon	68% lower	Median	68% upper
0	-4.949	-2.762	-0.172
6	-0.860	-0.008	0.832
12	0.251	0.521	0.705
24	-0.045	-0.005	0.040
36	-0.011	0.019	0.050

These credible sets summarize the arbitrary Haar weighting over rotations. They are not frequentist confidence sets for the identified set and, as we know, they need not cover the full range of sign-restriction-consistent responses.

(iii)

Figure 5 plots the identified-set bounds, posterior median, 68% credible set, and identified-set midpoint. The posterior median is not close to the center of the identified set at every horizon. The largest absolute gap between the median and midpoint is 1.059 annualized percentage points, occurring at horizon 0. For example, at horizon 12 the identified-set midpoint is 0.360, while the posterior median is 0.521. This illustrates that the truncated Haar prior supplies a weighting inside the identified set rather than a neutral estimate of its center.

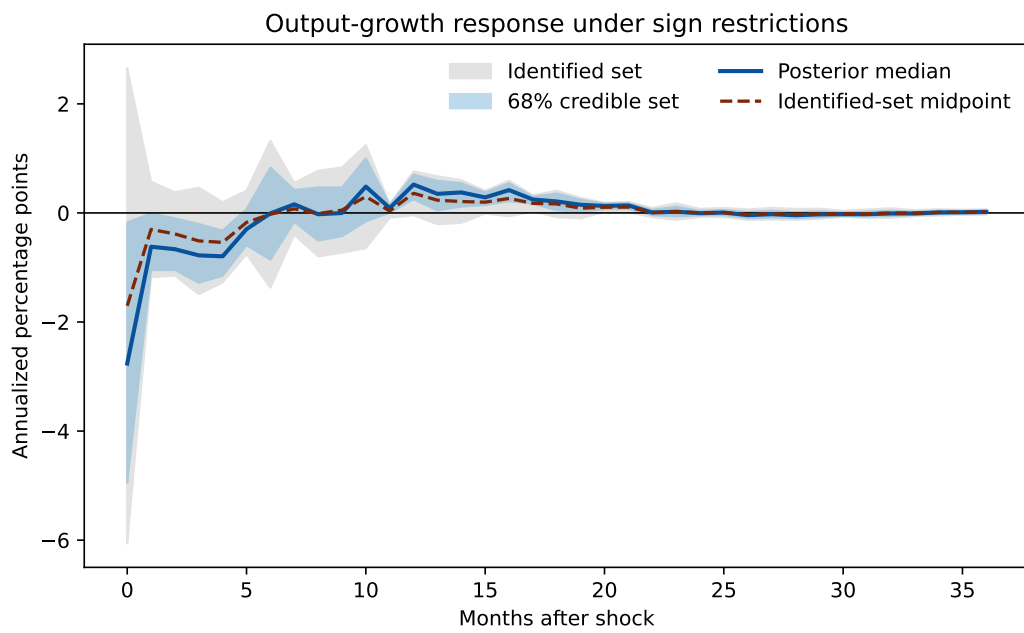


Figure 5: Output-growth response to a one-standard-deviation contractionary monetary policy shock under the sign restrictions that inflation falls and the real rate rises at horizons 0, 1, and 2.